

Matthew Tang

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EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Applied Science (BASc) in Electrical Engineering

Sep. 2026 – Apr. 2031

- **Awards:** President's Scholarship of Distinction, Roberts Onsite Engineering Entrance Scholarship

TECHNICAL SKILLS

Languages: C++, Python, Java, HTML, CSS, JavaScript

Hardware: Arduino, 3D Printing, Electrical Wiring

Software & Tools: Altium, Autodesk Fusion, VS Code, GitHub

DESIGN TEAM EXPERIENCE

Electrical Team Member

Sep. 2026 – Present

University of Waterloo Midnight Sun Solar Car Team

Waterloo, ON

- Designing the power tree for a multi-radio telemetry board (LoRa, Wi-Fi HaLow, 10BASE-T1S Ethernet) that streams live vehicle data to the chase car, converting a 12 V vehicle supply into 5 V, 3.3 V, and 1.8 V rails in Altium
- Built a component-level power budget from datasheets to size the buck converter and LDOs for a ~0.6 A worst-case load, flagging a ~1 W LDO thermal issue for design review
- Integrated 14 V TVS overvoltage protection and soft-start load switches so firmware can fully power down unused radios, targeting sub-0.1 mA standby draw

PROJECTS

Step-down Voltage Regulator PCB | *Altium*

- Designed a buck converter stepping a 10–15 V input down to a regulated 5 V/1 A output, calculating inductor and capacitor values from datasheet specifications
- Routed a 2-layer PCB with ground plane pours and design rule validation for manufacturability

Custom Pomodoro Timer | *C++, Arduino, 3D Printing, Autodesk Fusion*

- Engineered a device on Arduino Uno R3 with LCD screen and button usage for customizable work/break cycles
- Programmed timer logic and UI in C++, handling state transitions, button inputs, and audio/visual alerts
- Designed and 3D-printed a custom case in Autodesk Fusion, with a magnet-secured design for easy internal access

EXPERIENCE

Camp Leader

July 2024 – Aug. 2026

City of Toronto

Toronto, ON

- Facilitated a hands-on bridge-building challenge for 50 children ages 6 - 9, guiding participants through planning, construction, and testing while fostering creativity and problem-solving
- Conceived and led a cultural-awareness week, designing a Jeopardy-style trivia game, and organizing a multicultural potluck with 50+ participating families, while managing food-allergy compliance and safety
- Trained new staff and volunteers through structured on-site orientations, with peers shadowing to observe best practices

DECA Executive Entrepreneurship Trainer / Business Competition Coach Sep. 2024 – June 2026

Earl Haig Secondary School

Toronto, ON

- Designed and delivered a structured DECA training program (lesson plans, case studies) for 10 - 20 members
- Hosted direct feedback sessions before competition, resulting in 7 provincial qualifiers and 2 international qualifiers
- Represented Ontario at the International level in 2024 and 2025

Athletic Council Team Support Executive

June 2025 – May 2026

Earl Haig Secondary School

Toronto, ON

- Coordinated 5–10 volunteers across 10+ home games to manage logistics and equipment setup
- Produced 2 training videos on volunteer equipment use, mandated for all volunteer onboarding